The isotropic Schwarzschild relation ψ + Nψ = 2: a counting reading with one transfer postulate, and a closure inside matter

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

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Abstract

In isotropic coordinates the Schwarzschild metric satisfies ψ + Nψ = 2 for every mass, where N is the lapse and ψ the conformal factor of space. This record reads the two numbers of that relation from count vectors ("registers") with Dirichlet posteriors, and keeps apart what is computed, what is a premise of the author's framework, what is a choice, and one postulate. With five stated premises, and in the limit of many readings (corrections of order 1/T, measured): the lapse is the ratio of the Fisher steps per reading of two places, $N = \tau_0/\tau_x$; the depth is the normalised overlap of the two descriptions projected on the plane perpendicular to the line of sight, $\psi = 2\tau_x/(\tau_x + \tau_0)$; ψ/2 and Nψ/2 are then the shares of the place and of the distant observer in their joint count, and ψ + Nψ = 2 says that the two shares sum to one. The fragmentation of a register behaves as an isotropic Brownian motion in the three-dimensional Fisher space and deposits a measure $1/(2\pi r) - B$ with no free parameter. The conversion from that deposit to ψ is stated as a postulate (transfer of share, $\psi - 1 = \Sigma\, Gm/(2rc^2)$), chosen among four normalisations because it reproduces general relativity exactly: two alternatives are excluded by planetary ephemerides (β = 3/2 and 1/2), one differs from Schwarzschild only at third order. With it, one body gives the isotropic Schwarzschild metric (γ = β = 1) and two bodies …

A relation of gravity read as a count: the rule ψ + Nψ = 2 outside and inside stars

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Version 2.36.2 adds section 11. There the rule ψ + Nψ = 2 is read as an exchange. The exchange keeps the distant observer's unit of measurement fixed. It also shows which second-order term stays fixed. The deposit of section 3.6 is read as a probability of arrival. Version 2.36.2 also gives the precise reference for the formula $-\Phi/2c^2$ of sections 3.7 and 9. This form is in the first draft (preprint) of Verlinde's paper. The version published in the journal uses $-2\Phi/c^2$ instead. Version 2.36.1 had added sections 9 and 10.

MM (Mathematical Materiality) is the author's theoretical framework. In MM physical quantities are read from lists of counts. These lists are called registers. The registers are compared with each other. One measures how much they overlap in a statistical sense: these are the Bayesian overlaps. This work uses only a small part of MM. It is summarised in section 2.

1. The rule outside matter (in vacuum)

Now we look at an exact mathematical rule. It holds in the empty space around a mass.

In physics, a **metric** is the rule that says how distances and times are measured. Think of a map. The scale tells you how many real kilometres one centimetre on the sheet is worth. Near a star gravity changes both distances and the rate of

clocks.

Take a body that does not change in time: it is called **static**. It has the shape of a perfect sphere: it is called **spherically symmetric**. We use **isotropic coordinates**. They are a measuring grid that at every point behaves in the same way in all directions. It does not squash shapes.

In this grid the rulers are described by a spatial part: $\psi^4(dr^2 + r^2 d\Omega^2)$. Here **ψ** (psi) is a number. It is called the **conformal factor** or **depth**. It says how much space there really is in a stretch of the map, at one place, compared with a very distant place: near a mass, more rulers are needed to cover the same stretch. The rate of clocks is described by another number, **N**. It is called the **lapse**, that is, the rate. It says how slow a clock runs at that place, compared with a distant clock. For convenience physicists set the gravitational constant and the speed of light equal to 1 (G = c = 1).

In Einstein's general relativity two formulas hold. They hold in the empty space outside a mass M, at distance r. The first is $\psi = 1 + M/2r$. The second concerns the product of the two numbers: $N\psi = 1 - M/2r$. The two expressions are **harmonic**. A harmonic function is a smooth distribution. It does not pile up at any empty point. Think of the temperature of a room with no heaters switched on in the middle of the air.

Let us add ψ and Nψ. The two pieces +M/2r and −M/2r cancel. What remains is:

$\psi + N\psi = 2$

This relation holds exactly for the **Schwarzschild metric**. It is Einstein's solution for the empty space around a spherical mass. Relativity measures the total mass from far away in two ways. They are called the monopoles M_ADM and M_Komar. Here they give the same value M: M_ADM = M_Komar = M.

The converse also holds, for **one centre**. It is checked with an exact formula. Take $\psi = 1 + a/r$. Impose the rate $N = 2/\psi - 1$, which is another way of writing $\psi + N\psi = 2$. We find exactly the isotropic Schwarzschild metric, with mass M = 2a. In the Solar System two parameters are used to test models of gravity. They are called **γ** (gamma) and **β** (beta). In this exterior space both are exactly 1: $\gamma = \beta = 1$.

The rule, however, does not extend to several masses, when the spherical shape is lost. Let us try with two masses, at the points x_1 and x_2. We write $\psi = 1 + a_1/|x - x_1| + a_2/|x - x_2|$. This ψ is still harmonic. But now we impose the static rate $N = 2/\psi - 1$. Then Einstein's vacuum equations are no longer satisfied. Curvature is measured with a mathematical object: the **Ricci tensor**. In vacuum it should be zero. Here instead it is not zero. A residue remains, proportional to the product of the two masses (of order $a_1 a_2$). The residue is there in an open region of space. It disappears only if we remove one of the two masses. Technical note: this two-centre ψ is the Brill–Lindquist conformal factor. In general relativity it describes two black holes at rest, at a single initial instant that is symmetric in time. What falls here is something else: the reading with the static rate $N = 2/\psi - 1$ kept fixed in time.

The relation $\psi + N\psi = 2$ therefore holds only in exact spherical symmetry. **Domain of this work: static, spherically symmetric, one body.**

2. The model in five lines

Before the calculations let us look at the five building blocks of the model. With these the model builds space starting from counts.

1. **The register.** A **register** is a list of counts $c \in \mathbb{N}^K$. It is divided into K = 4 categories. Each category k has at least one count ($c_k \geq 1$). The sum of all the counts is n. A concrete example: a notebook with 4 columns. In the columns you mark with a cross the colour of the cars that go by: red, blue, green, yellow. The state of the register is described by a probability distribution. It is called the **Dirichlet posterior distribution**. It has a value of **evidence** $E(c) = \log[(K-1)! \prod_k c_k! / (\Sigma_k c_k + K - 1)!]$. The evidence measures how probable it is to observe exactly that list of counts.

1. **Sampling and Fisher distance.** The registers are drawn at random in a uniform way. The **Fisher measure** is used on the space of probabilities, which is called the simplex (Amari and Nagaoka 2000). In practice 4 random numbers are generated with a bell curve (Gaussian g). Then $p = g^2/|g|^2$ is computed. Then the counts are drawn with a rule called multinomial: $c = 1 + \text{Multinomial}(n - K, p)$. The distance between two registers is not measured with a ruler. It is measured with the **Fisher distance**: $d(p, p') = 2 \arccos \Sigma_k \sqrt{(p_k p'_k)}$. Two registers are close if they have similar percentages in the 4 columns. They are far apart if they are very easy to tell apart. The idea of using statistical distinguishability as a distance goes back to Wootters (1981). The idea that information geometry is the geometry of physical space is Caticha's (2005, 2016, 2019; see section 9).

1. **Links between registers.** Two registers i and j are connected by a weight. It is called the **normalised Bayesian overlap**: w_ij = exp[E(c_i + c_j) − ½E(2c_i) − ½E(2c_j)]. The weight measures how well the two registers can fit in one single count. Each register has weight 1 with itself (self-weight 1). Only the links between nearby registers are kept. Nearby means within 3.5 typical widths of fluctuation. One width is 2/√n in the coordinates 2√(c/n).

1. **The operator on the graph.** The connected registers form a network. It is called a **graph**, like a network of towns joined by roads. On the network a mathematical operator is defined. It measures how values change from a node to its neighbours. It is the **normalised graph Laplacian** L = I − D^(−1/2) a D^(−1/2). Here W is the table of the weights w_ij. The d are the row sums of W. Then a = $d^{-1}Wd^{-1}$. The D are the row sums of a.

1. **The static response.** We place a point source δ at the most uniform register. It is the one with the 4 columns almost equal. We look for the response that stays still in time. The equation L G = δ is solved inside a sphere of radius 0.7 around the source: the Fisher ball. Outside this radius we set G = 0. The value 0.7 is a numerical choice. The potential that comes out is φ = D^(−1/2) G.

This graph lives in a three-dimensional space: the space of register states. Why three dimensions? The categories are K = 4. But if you know the first three percentages, the fourth is fixed. The total, in fact, must be 100 %. So 4 − 1 = 3 degrees of freedom remain. In this abstract space there are distances. Identifying them with the distances of physical space is a **premise** of the framework. It is premise P3 of section 3. It is not a result proved by calculation.

Section 3 adds three concepts of the same framework. They are the effective density of a register, the image of a register seen along a line and the fragmentation of a register. Section 7 uses a separate test bench, described there.

3. From counting premises to the numbers ψ and N

Now we see how the two numbers of gravity are read from the count registers. They are ψ, the spatial depth, and N, the rate of clocks.

To be clear, every step has a label in square brackets:

- **[C]** marks what is **computed**, by hand or by computer;
- **[P]** marks a **premise**, that is, a starting point assumed by the theoretical framework;
- **[S]** marks a **choice** made among various possible alternatives;
- **[T]** marks the single **transfer postulate**, which joins the counts to gravity.

This whole chain of steps is static: it does not change in time. Its laws hold when the number of readings T is very large. This is called the many-readings limit. There are small corrections, proportional to 1/T.

3.1 The elementary identity [C]

Let us start with a purely mathematical property. It concerns two bell curves. Imagine two blurred spots of light, with the same centre. They are in a space of D dimensions. They are described by two **isotropic Gaussian** distributions, p_0 and p_x.

- The first has variance s_0^2. The variance is the width of the blur, squared. It represents the resolution of the **distant observer**.
- The second has variance s_x^2. It represents the resolution **at the place** observed.

The overlap between two spots is the integral of their product: Λ_ab = ∫ p_a p_b. The normalised overlap seen from b towards a is ℓ(a → b) = Λ_ab/Λ_bb. We call $q = s_0^2/s_x^2$ the ratio between the two widths.

The calculation gives:

- ℓ(0 → x) = (2/(1+q))^(D/2)
- ℓ(x → 0) = (2q/(1+q))^(D/2)

Something remarkable happens when the dimensions are **D = 2**, that is, on a plane. The exponent D/2 becomes 2/2 = 1. Then the sum of the two overlaps is: 2/(1+q) + 2q/(1+q) = 2(1+q)/(1+q) = **2**, for any value of q.

There is another very common statistical tool. It is called **relative entropy**, or Kullback–Leibler (KL) divergence. With

this tool the sum of the two directions does not give a constant: KL(0‖x) + KL(x‖0) = (D/2)(q − 1)²/q. The rest of the section says why the two overlaps ℓ correspond to the depth ψ and to the product Nψ.

3.2 The five premises [P]

Here are the axioms of the MM framework. Axioms are the basic principles, not proved. Here they are written in a clear form.

- **P1 (clock).** Again and again a new **reading** arrives, that is, a new count. Each time the register of a place is updated a little. The **proper time** of that place is the time marked by a clock that stays there. At each reading it advances by an amount: the Fisher length of this small update. Readings arrive everywhere at the same rate. Measuring time with the size of statistical fluctuations is an idea of Caticha from 2011: entropic time. Ipek, Abedi and Caticha made it local in 2018, and Ipek and Caticha in 2020. In their works, however, the rate N is already given by the geometry. It is the rate that sets the fluctuations. Here the direction is reversed: it is the counts that set the rate N. We will see this in section 9.

- **P2 (image).** There are two points of view: the place near the mass and the distant observer. They describe the same thing: a register with the same composition in percentages. But they have collected a different number of readings. They compare their descriptions by looking at the **image**. The image is projected on a plane perpendicular to the line of sight that joins them. Think of a ball seen from far away: you see a flat disc. So this comparison takes place in **two dimensions** (D = 2).

- **P3 (distance).** Physical distance in space is the measure of the **fragmentation** of a register. Fragmentation says how much the copies of the register have moved and spread in the space of register states.

- **P4 (mass).** The number of copies of a register, that is, its multiple, is the **mass** of the body. The effect of several copies adds up.

- **P5 (absorption).** A copy arrives at a place. There it is absorbed into the joint state, that is, the united state, of the two points of view. This joint state stays complete: the total of the parts goes on making the whole.

3.3 The clock: $N = \tau_0/\tau_x$ [C]

Now let us see why a clock with more counts beats time more slowly.

The **effective density** of a register is the number τ = T + 1. T stands for the readings collected. The + 1 accounts for the starting counts (prior pseudo-counts). A new reading arrives: by how much does the state of the register move on average? Think of a tank of water. If there is only one litre in the tank, one more glass changes the level a lot. This is the case with few counts. If there are a thousand litres, the same glass moves the level very little. This is the case with many counts.

Let us compute the average step of a new reading, in the local Fisher metric. It is the root-mean-square Fisher step. It is exactly $\sqrt{(K-1)/\tau}$, for any percentage composition of the register. For K = 4 categories the calculation with formulas is exact. In computer simulations it holds within 0.23 %. It is the same for all compositions within 0.35 %. The finite Fisher –Rao distance also agrees, when every category contains many readings.

Let us compare the clock step at place x with that of the distant observer 0. Place x has density τ_x. The distant observer has density τ_0. The ratio between the two steps is $\boldsymbol{\tau_0/\tau_x}$. It is checked by computer within 0.3 %. For equal composition, the ratio between the variances of the Dirichlet distributions also gives exactly the same result.

Now we use premise P1. The rate of the place's clock, compared with the distant one, is the lapse N. And it is exactly this ratio:

$$\mathbf{N = \tau_0/\tau_x}$$

Where there are more readings, τ_x is larger. Then the step for each reading is smaller. And the clock of the place runs slower (N < 1). Let us look at the form of this law. The rate N is the ratio between a quantity measured far away and the same quantity measured at the place. The form is identical to the famous **Tolman–Ehrenfest relation** of the physics of heat: T_loc N = T_∞ (Tolman and Ehrenfest 1930). Here, in place of the temperature T, there is the density of readings τ.

3.4 The depth: $\psi = 2\tau_x/(\tau_x + \tau_0)$ [C, many-readings limit]

Now we compute the second number, ψ. We do it by comparing the images of the two registers on a flat screen.

We take the probability distributions Dir(τp) of the two registers. We use the Fisher coordinates $y = 2\sqrt{p}$. We project them on the plane perpendicular to the line of sight between the place and the distant observer. Premise P2 asks for this. Then we compute the normalised overlap seen from the place towards the distant observer: $L\perp(x \to 0) = \int f_x f_0 / \int f_0^2$.

The calculation was made on the true projected Dirichlet images. They were sampled by computer, without the approximate Gaussian formula. The result coincides with the formula $\mathbf{2\tau_x/(\tau_x + \tau_0)}$:

- within 0.44 % when the distant observer has $T_0 = 100$ readings;
- within 0.10 % at $T_0 = 300$;
- within 0.03-0.04 % at $T_0 = 1000$-3000.

The difference is a small correction of order 1/T. The result does not depend on the direction of the line of sight, within 0.1 %. There is more. The ratio between the two opposite directions, $\int f_0^2/\int f_x^2$, tends to τ_0/τ_x. So we find again the same rate N of section 3.3, by a second independent route. And what if we do not project on a plane? What if we make the overlap in the whole three-dimensional space of states? Then the formula changes: it goes with the exponent 3/2 instead of 1.

[S] Declared choice. We identify the spatial depth **ψ** with the directional overlap seen from the place: $\mathbf{\psi = 2\tau_x/(\tau_x + \tau_0)}$. Why not a symmetric formula, which treats the two points of view in the same way? Because a symmetric comparison does not distinguish who has more readings from who has fewer. In mathematics it is an even function of the relative difference $z = (\tau_x - \tau_0)/(\tau_x + \tau_0)$. With a symmetric formula, at first order we would see no curvature of space. We would get γ = 0. It is a reading that falls, because observations contradict it. So the directional overlap is chosen. It responds linearly to the excess of readings: it distinguishes the plus sign from the minus sign.

3.5 Two shares of one joint count [C]

Now let us put N and ψ together. We see what simple meaning the rule $\psi + N\psi = 2$ takes.

We stay in the many-readings limit seen in 3.4. We divide our two numbers by 2:

- $\mathbf{\psi/2 = \tau_x/(\tau_x + \tau_0)}$ is the fraction of readings of place x. It is its **share** of the joint total of the two points of view $(\tau_x + \tau_0)$.
- $\mathbf{N\psi/2 = (\tau_0/\tau_x) \cdot \tau_x/(\tau_x + \tau_0) = \tau_0/(\tau_x + \tau_0)}$ is the fraction of readings of the distant observer 0. It is its **share** of the same total.

An example with made-up numbers. Two kids put their stickers together in a shared album. The first has put in 60 % of the stickers ($\psi/2 = 0.60$). Then the second has put in the remaining 40 % ($N\psi/2 = 0.40$). The sum of the two shares is 1, that is, 100 % of the album. Let us multiply by 2. **$\psi + N\psi = 2$ says only this: the two shares sum to one.**

This is the reading proposed for the Schwarzschild relation. It holds outside matter and inside matter (section 4). In section 11 we will also read it as the conservation law of an exchange. In isotropic coordinates the geometric shape of space is flat. All the information on the metric of space is in the scale factor ψ. It is exactly the local scale that information geometry, on its own, left free (Caticha 2016; section 9).

3.6 The source: fragmentation gives the 1/r behaviour [C]

So far we have talked about the density of readings τ_x at a place. But why are there more readings near a mass? And why does this excess fall off exactly as **1/r**, the inverse of the distance?

We follow the **fragmentation** of a register with K = 4 categories. Its copies move away. They lose precision, reading by reading. This process behaves exactly like an **isotropic Brownian motion** in the three-dimensional Fisher space. A Brownian motion is a random walk, in small steps, in all directions. Think of a grain of pollen hit by water molecules. The internal clock s of this walk is the sum of the squares of the Fisher steps.

The test uses 60,000 chains simulated by computer. The motion is the same in all directions (isotropic) within 1.8-5 %. The variance in each direction is equal to s within 1.4 %.

At each step we weight what the copy leaves at the place. The weight is the precision lost, **Δs**. Then the quantity

deposited at distance r from the source follows the law **1/(2πr) − B**. Once the conventions of the calculation are fixed, there are no parameters to adjust by hand. The conventions are: Fisher coordinates, clock s, tangent Euclidean volume, minimum stopping threshold n_min. This geometric calculation does not give the value of G in physical units, such as metres and kilograms. Between the distance r = 0.005 and r = 0.04 the simulated points are compared with the formula. The mean ratio is **0.998**. No distance band deviates by more than 3 %. The constant B arises because fragmentation stops when the counts run out. This happens at a maximum time s_max. The whole profile is described by the function $[1/(2\pi r)]\ \mathrm{erfc}(r/\sqrt{(2s_{max})})$.

In mathematics $1/(2\pi r)$ is the **Green function** of Brownian motion in three dimensions. It is the average trace left by a random walk in 3D space. The MM framework adds one thing: the statistical fragmentation of a 4-category register behaves exactly like that Brownian motion, with that precise step.

[S] Declared choices. Two things are declared choices: the way the register loses counts and the weight Δs given to the deposit. In an earlier test each reading had the same weight, without Δs. That test missed the registered slope 1/r, and that reading fell. With premise P3, the distance r in the space of registers is the physical distance. With premise P4, n copies deposit n times as much.

3.7 The junction with physics: Postulate T [T]

Now we have the pieces. The final bridge is missing. In steps 3.3-3.6 nothing says how to convert the measure deposited by fragmentation into the physical number ψ, with the constant G. This conversion is openly declared as a postulate.

> **Postulate T (transfer).** Each deposit moves a share of count from the distant observer to the place: $\psi - 1 = \Sigma_i w_i$. For each mass m_i at distance r_i the share moved is $w_i = Gm_i/(2r_ic^2)$. The coefficient is calibrated by experiment: the constant G is measured, not computed. The shares always stay positive ($\Sigma_i w_i < 1$). For a single body of mass M one must stay outside the radius $r > GM/(2c^2)$.

Postulate T is motivated by premise P5. An absorbed copy, in fact, redistributes the parts of a joint state that stays complete. But Postulate T does not necessarily follow from P5. Adding the counts and recomputing the total would also give a complete state.

The quantity moved by the postulate is **$w = GM/(2rc^2)$**, written with the isotropic radius r. At first order it coincides with **$-\Phi/2c^2$**. First order means far from the mass, in a weak field. Here $\Phi = -GM/r$ is Newton's gravitational potential.

Erik Verlinde wrote a famous paper on emergent gravity. In the first version (preprint: arXiv:1001.0785v1 of 2010, equation 3.16 and section 4) there is exactly this quantity $-\Phi/2c^2$. There it measures the loss of detail for each bit of information. This is called *coarse graining*. It is a number between 0 and 1. It is 1 only on the event horizon of a black hole.

In the version published in the journal (JHEP 04 (2011) 029, equation 3.12 and section 4) Verlinde writes the same thing as $-2\Phi/c^2$. This is 1 on the horizon with the standard coordinates, called areal coordinates. In those coordinates the radius of the horizon is $r = 2GM/c^2$. In our isotropic coordinates, instead, the black-hole horizon lies at **$r = GM/(2c^2)$**. And it is exactly there that our share w reaches 1. Then ψ = 2 and N = 0: the clock is stopped. This precise value comes from a choice: Postulate T is written with the isotropic radius. It does not come from an identity with Verlinde's formula beyond the first approximation.

Why, in Postulate T, did we choose exactly ψ − 1 as the measure of the excess of readings? The excess of readings ($\tau_x - \tau_0$) can be compared with a reference value in four natural ways. We ask the same thing of all of them: at large distances they must give Newton's gravity. That is, we take them proportional to $u = GM/(rc^2)$. The table says what happens with each of the four choices.

Choice: excess of readings divided by…	Parameter β	Parameter γ	Third order of N	Comparison with astronomical data
S1 the distant count: $(\tau_x - \tau_0)/\tau_0$	3/2	1	$-u^3$	**The reading falls (excluded).** The measurements (Will 2014) say that β − 1 is below about 10^{-4}. Here instead β − 1 = 1/2.
S2 the count of the place: $(\tau_x - \tau_0)/\tau_x$	1/2	1	0	**The reading falls (excluded).** Here too β is completely outside the measurements of the perihelion of the planets.
S3 the geometric mean: $(\tau_x - \tau_0)/\sqrt{(\tau_x\tau_0)}$	1	1	$-u^3/8$	**Passes today's tests.** It gives γ = β = 1. But at third order it deviates from

				Schwarzschild: $-u^3/8$ instead of $-u^3/4$.
S4 the joint count: $(\tau_x - \tau_0)/(\tau_x + \tau_0) = \psi - 1$	1	1	$-u^3/4$	**Exact isotropic Schwarzschild**, at every order.

Choice **S4** is adopted. The reason: it reproduces general relativity exactly at all orders. It is an external criterion: the correspondence with Einstein's theory. It is not a choice imposed by the premises of MM. Choice S3 would give observable differences only in extremely strong gravitational fields.

3.8 What follows from Postulate T [C]

Let us see the direct consequences of the postulate. First for a single body, then for two bodies.

- **A single body.** We have $\psi = 1 + GM/(2rc^2)$. The shares are complete (3.5), so $N = 2/\psi - 1$. This is exactly the isotropic Schwarzschild metric. $\gamma = \beta = 1$ holds. Outside the mass the orbits are those of general relativity (the program perielio.py computes them). For example, the shift of Mercury's perihelion comes out the same.

- **Two bodies.** The sum $\psi = 1 + \Sigma_i Gm_i/(2r_ic^2)$ exactly satisfies one of Einstein's vacuum equations. It is called the **Hamiltonian constraint**. It governs the curvature of space at a given instant. These are the famous initial data of **Brill and Lindquist (1963)**: two black holes at rest for an instant. Important note: with reading S1 the constraint would not be satisfied. S1 adds the raw counts instead of the shares.

- **Two bodies do not stay still.** Let us extend the rate rule $N = 2/\psi - 1$ to the case of two bodies. Then the full static vacuum equations are not satisfied. The spatial tensor $E_{ij} = R_{ij} - D_iD_jN/N$ is not zero. In physics this is as it should be. In general relativity too, two masses left at rest do not stay at rest. They start to fall towards each other. Two bodies at rest are therefore only an initial snapshot. Then everything must change in time. Here E_{ij} is not directly a measured force. And our static chain does not yet contain the laws of motion, that is, the dynamics.

3.9 Taking stock of section 3

Let us sum up honestly what has been done and what has not. The Schwarzschild metric **is not derived** from scratch in this work. It enters through Postulate T. It enters together with the 1/r behaviour of fragmentation (3.6) and the correspondence choice S4. The counting chain really gives three things:

1. the mathematical form of the two quantities: N as a ratio of densities, ψ as a share;
2. the relation $\psi + N\psi = 2$, read like this: two shares of a total sum to one;
3. the 1/r law, produced by the Brownian fragmentation of the registers.

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4. Inside matter: a phenomenological closure, not yet a theory

So far we have stayed in empty space, outside the star. Now we go **inside the matter** of the star. Let us see what changes if the rule $\psi + N\psi = 2$ holds there too.

In the reading of section 3, the relation says one thing: the two shares of the joint count sum to one (3.5). Keeping it valid inside matter means imposing this completeness **point by point**. Even in the middle of the star's gas.

Let us first see what ordinary general relativity does. Take a static star, in isotropic coordinates. There are two main equations. In them ρ is the mass-energy density of matter and p is the pressure of the fluid.

1. The **Hamiltonian constraint**, for space: $\nabla^2\psi = -2\pi\rho\psi^5$.
2. The **lapse equation**, for time: $\nabla^2(N\psi) = 2\pi N\psi^5(\rho + 6p)$.

The symbol ∇^2 is called the Laplacian. It measures how much a value at a point deviates from the average of the nearby points. Let us add the two Einstein equations. We get:

$$\nabla^2(\psi + N\psi) = 2\pi\psi^5[N(\rho + 6p) - \rho]$$

Inside the star this result is not zero. It is a small but present quantity: it is of first post-Newtonian order. Now let us add

up this effect over the whole volume of the star. In mathematics this is called taking the volume integral. By a theorem (the divergence theorem) the result is a flux through the surface. It depends only on how ψ and N behave far away: $\psi \to 1 + M_ADM/2r$ and $N \to 1 - M_Komar/r$. One gets:

$$\int \nabla^2(\psi + N\psi)\, d^3x = 4\pi(M_Komar - M_ADM)$$

In general relativity, for every static star, the two total masses M_Komar and M_ADM are exactly equal. Beig proved this in 1978; see also Tolman 1930. So this total integral is **zero**. In simple words: in general relativity the sum $\psi + N\psi$ inside the star does not stay at 2 point by point. But added up over the whole star, the effect cancels. Keeping $\psi + N\psi = 2$ inside matter means imposing **point by point** what general relativity guarantees only **as a whole**, over the whole star.

The choice of variable is an assumption

The rule is written for the isotropic factor ψ. That is, the metric of space has the form $\psi^4\delta_ij$. This is a precise choice: an **assumption**. With the usual areal coordinates, the similar property in vacuum would be $g_tt\, g_rr = -1$. Jacobson studied it in 2007. Imposing that formula inside matter would give a different result.

What is replaced and what it costs

Here is how our model works inside a star:

1. It **keeps** the Hamiltonian constraint: $\nabla^2\psi = -2\pi\rho\psi^5$.
2. It **replaces** Einstein's lapse equation with an algebraic rule: $N = 2/\psi - 1$.
3. It **takes the pressure** of the fluid from the law of conservation of energy and momentum. This law says that every layer of the star must stay in balance without collapsing: $dp/dr = -(\rho + p)\, d \ln N/dr$.

What is the price of this move? **The pressure of the fluid no longer enters the equations that generate the gravitational field.** Inside matter the geometry is no longer generated entirely by the fluid, as Einstein's equations require. For this reason, inside matter, the proposal is a **phenomenological closure** of the static equations. A phenomenological closure is a practical rule. It closes the system of equations and makes it possible to do the calculations. **It is not yet a complete theory.**

Let us see the numbers for an ideal star, with uniform density: ρ the same at every point.

- The part of the Einstein tensor that concerns time gives exactly the density: $-G\hat{}t_t/8\pi = \rho$.
- The parts that concern space, instead, do not give the pressure of the fluid p. At the centre of the star the geometry *sees* about half of the fluid pressure: $G\hat{}r_r/8\pi = G\hat{}\theta_\theta/8\pi \approx$ **0.50 p_c** in the weak field.
- Going towards the surface, the difference becomes **anisotropic**. That is, it changes with direction. It is different in the radial direction, outwards, and in the tangential direction, along the surface. The tangential part is positive.

A relativist physicist reads this star with Einstein's equations. Then they say: inside the star, besides the normal fluid, there is an **extra anisotropic stress**. This stress is conserved on its own. And it has an energy density equal to zero.

This touches a famous limit of the physics of stars. How compact can a star be before it collapses into a black hole? **Compactness** is measured with the ratio $C = GM/(Rc^2)$. Here M is the mass and R the radius of the star.

- In 1959 **Buchdahl** proved a limit. It holds in general relativity, for an isotropic perfect fluid with density that does not increase outwards. Compactness can never exceed $C =$ **4/9 ≈ 0.4444**. Otherwise the pressure at the centre would become infinite.
- In 2008 **Andréasson** proved a more general theorem. It holds for any static spherical matter, even anisotropic. Anisotropic spheres had been studied by Bowers and Liang in 1974. It always holds that $2m/r \le 8/9$, that is, $C \le 4/9$. Two conditions on energy are needed, though. The radial pressure must not be negative: $p_r \ge 0$. The sum of the pressures must not exceed the density: $p_r + 2p_t \le \rho$.

So let us take a star of our model with compactness above 4/9. If we read it in general relativity, it must violate one of Andréasson's two conditions. It violates it through the extra anisotropic stress. Inside matter, then, this model **is not** general relativity plus a simple extra hypothesis.

The maximum compactness limit in one formula

Take a star with uniform proper density. The Hamiltonian constraint becomes an equation known in astrophysics. It is the **Lane–Emden equation with index n = 5**. It is solved exactly by the formula $\psi = \psi_c(1 + br^2)\hat{}(-1/2)$. Here ψ_c is

the value at the centre. At the surface (r = R) we match the interior with the empty exterior space. One finds ψ_R = ψ_c^(2/3).

The family of possible stars ends when the centre reaches the maximum of the shares: **ψ_c = 2**. This means N_c = 0: time is stopped at the centre of the star. There the maximum compactness is exactly:

C_max = GM/(Rc²) = 2^(1/3) − 2^(−1/3) ≈ 0.4662, against the Buchdahl limit **4/9 ≈ 0.4444**.

What happens at the compactness of real stars

There is no need to reach the extreme limit to see a difference. Astronomers observe neutron stars with typical compactness **C = 0.16, 0.19 and 0.22**. Take a uniform-density star with these compactness values. The central pressure that our model needs to hold up the star is **lower** than that of general relativity. It is lower by **7.4 %, 9.3 % and 11.4 %**, respectively.

Of course, a uniform density, the same from the centre to the crust, is not realistic for a real neutron star. One can use a realistic equation of state of nuclear matter. Then the curve that links the mass to the radius of the star would shift. Computing this shift is the next step of the work. It is not done here yet. Other density profiles, which fall from the centre outwards, are studied in the separate deposit **MM 2.34**, by the same author, on Zenodo. The energy conditions are also studied there.

5. All the assumptions and how they can be put to the test

So as to hide nothing, we collect in a table all the assumptions of the model. For each one we state its logical status. And we state which observation or calculation would make it fall.

#	Assumption	Status in the model	What would make it fall
1	**Source:** $\psi - 1 = \Sigma_i w_i$, with $w_i = Gm_i/(2r_ic^2)$ (Postulate T, section 3.7)	Postulate. It is chosen among four normalisations, to have exact agreement with general relativity.	A strong-field measurement in favour of choice S3, which changes the metric at third order.
2	**Harmonic propagation:** the deposit falls off as 1/r	Fragmentation (3.6) and the graph (section 6) give 1/r in the space of register states. Identifying it with physical space is premise P3.	A free exponent on the graph that stayed below 1 even with a finer resolution and more random seeds.
3	**The rule ψ + Nψ = 2 everywhere**, also inside matter. There it takes the place of the lapse equation of general relativity.	Assumed. In the reading of 3.5 it is the completeness of the two shares, kept point by point. In the reading of 11.2 it is the conservation of an exchange. Outside matter it is a property of general relativity.	A measurement of the mass and radius of a neutron star in favour of general relativity, after using a realistic equation of state. The value 0.466 is the maximum only for uniform density. For densities that fall outwards the maximum is lower (MM 2.34).
4	**Condition far from the mass:** $\psi \to 1$ at infinity. It fixes the unit of measurement of the distant observer.	Assumed.	—. If we set for example ψ = 3/2 + a/r, the Einstein tensor in vacuum would not be zero.
4′	**The chosen variable** is the isotropic conformal factor: metric of space $\psi^4\delta_{ij}$.	Assumed. In areal coordinates the similar property in vacuum is $g_{tt}\,g_{rr} = -1$ (Jacobson 2007).	—
5	**Form of the metric:** $g_{tt} = -N^2$, $g_{ij} = \psi^4\delta_{ij}$. Test particles follow the straightest lines (geodesics) with energy $H = N\sqrt{(m^2 + p^2/\psi^4)}$.	Assumed.	A measurement of time or distance that does not follow this geometric form.
6	**Universality of the rate N.** It is the equivalence principle: all bodies fall in the same way.	Assumed.	A free-fall experiment that depended on the material. Today we know that the fall is the same for all materials up to one part in 10^{15}, that is, 10^{-15}.
7	**The constant G** is calibrated by a measurement. It is assumed that the parameter a is proportional to the mass.	Assumed.	A constant G that changed from one body to another.
P1-	**Premises of the framework** (section 3.2)	Starting axioms.	—

P5			
P6-P7	**Premises of the moving clock** (section 10): second link, no place assigned in advance.	Starting axioms.	—
P8	**Premise of the exchange** (section 11.2)	Starting axiom.	—
—	**Section 11.2:** the two shares as parts that exchange with each other. Their sum is conserved. Light reads the product of the local units of time and length.	Declared formalisation of premise P8.	—
—	**Moving clock:** the clock reads the number of copies. It combines the two opposite directions of light by multiplying them (section 10).	Declared choice.	—
—	**Choices of section 3:** effective density τ = T + 1; directional overlap identified with ψ; loss mode and weight Δs in the fragmentation; same composition of the two points of view.	Declared choices (sections 3.3-3.6).	—
—	**Many-readings limit:** the laws of sections 3.3-3.5 hold with small corrections of order 1/T.	Measured in the tests: deviation of 0.44 % at T = 100 readings and of 0.03 % at T = 3000.	—

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6. The 1/r behaviour checked on the model's graphs

Now we build on the computer the network of registers of section 2. Then we measure how the potential φ falls off far from the source.

We compute the average values of φ on spherical shells, at various distances r. To these values we fit a curve of the type A r^(−p) + B. The exponent **p** is left free. So we see whether it really tends to 1, that is, to r^(−1) = 1/r. We made three trials. Each time we increased the number n of counts per register:

1. with **n = 512** counts per register, on a network of **64,000** registers, the free exponent is **p = 0.937**;
2. with **n = 1024** counts per register (**181,019** registers) the exponent rises to **p = 0.956**;
3. with **n = 2048** counts per register (**512,000** registers) the exponent reaches **p = 0.980**.

These are three points. Each one is computed with a single random seed. So there are no statistical error bars.

This approach to 1 is exactly what mathematics predicts. It is proved by Belkin and Niyogi (2008) and by Coifman and Lafon (2006). Take Laplacian operators on a graph, with links built from bell-shaped weights, similar to Gaussians. When the points increase, these operators approach the continuous geometric operator of the space beneath them. It is the Laplace–Beltrami operator. The state space of a register with K = 4 categories has **three dimensions**. Therefore its static response must tend to 1/r.

The 1/r behaviour therefore reflects the fact that K = 4. It is not the result of a dynamical law of motion. Section 3.6 had given a second route to 1/r. It started from the fragmentation of a single register. Both routes rest on the three dimensions of the state space for K = 4. Premise P3 identifies this space with physical space.

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7. The parameter γ on a ring with finite counts

Careful: in this section there are two statements. They concern two different mathematical objects.

- Take the continuous exterior metric, the one with infinite counts. There the post-Newtonian parameter γ is **exactly γ = 1** (section 1).
- Then there is a separate test bench, with a finite number of counts. It is a ring of registers, linked with the square

of the overlaps. Here we ask how large the small deviation from 1 is, due to the fact that n is finite.

We take as given the relation **γ = 2p − 1**. It comes from the author's private working files. It is not part of this deposit and cannot be checked here. In this formula p is the exponent with which the slowest vibration of the ring changes with the resolution. With this relation the simulations find:

γ − 1 ≈ 7.9 / n

Today the most precise measurement of γ in the Solar System comes from the **Cassini** probe. These are the radio signals sent by the probe near Saturn (Bertotti, Iess and Tortora 2003). The measurement is $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$. The uncertainty is $\mathbf{2.3 \times 10^{-5}}$. The deviation 7.9/n must stay below Cassini's uncertainty. Then each register must contain at least $\mathbf{n > 3.4 \times 10^{5}}$ counts, that is, more than three hundred and forty thousand. This estimate was found after the registered test was done.

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8. What is not claimed in this work

To avoid misunderstandings or exaggerations, here is the list of what this work does **not** claim to do.

- **It is not claimed that the Schwarzschild metric is derived from scratch.** Outside matter the relation ψ + Nψ = 2 is a known property of general relativity. In the chain of section 3 it enters through Postulate T and choice S4.
- **It is not claimed that the premises P1-P5 are proven results.** They are the starting axioms of the theoretical framework. All the choices of section 3 are openly declared.
- **It is not claimed that the laws of sections 3.3-3.5 are exact with few counts.** They hold in the many-readings limit, with corrections of order 1/T.
- **It is not claimed that the 1/r behaviour comes from a dynamical force.** It is the Green function of a Brownian motion in three dimensions (3.6). And it is the expected mathematical behaviour of the graph Laplacian in three dimensions (section 6).
- **It is not claimed to predict the numerical value of G.** The gravitational constant is measured by experiments.
- **It is not claimed that 0.466 is the maximum compactness limit for any star.** It is the maximum only in the ideal case of uniform density.
- **It is not claimed that the model reproduces general relativity inside matter.** As it is built, it does not. Inside matter it is a phenomenological closure of the static equations, not yet a complete theory.
- **No dynamics is proposed**, that is, a law of evolution in time. And no general covariant formulation is proposed. Two bodies at rest are only a snapshot at the initial instant. They are not a solution that stays still.
- **It is not claimed that the ideas of section 9 are all new.** They belong to the earlier works cited there. There it is stated precisely what this deposit adds of its own.
- **It is not claimed that the mathematics of the moving clock (section 10) is new.** It coincides with causal-set theory (*causal sets*) and with Bondi's k-calculus. And it is not yet joined in a single chain with the gravitational rate N of section 3.
- **It is not claimed that section 11 proves ψ + Nψ = 2, Postulate T, G or the horizon radius GM/(2c²) from scratch.** Section 11 gives the relation a physical reading as an exchange. And it gives the deposit the meaning of a probability of arrival. Nor is it claimed that the fragmentation test bench rigorously proves the Brownian limit of the first arrival. The numerical results are compatible with that hypothesis.

Other lines of research of the author are not part of this deposit.

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9. Comparison with the work of other scientists (added in version 2.36.1)

The chain of section 3 uses several ideas already introduced by other physicists. The table shows, for each ingredient, what was already there and what changes here.

Element of this work	Earlier work	What is different here
Distance as Fisher distinguishability (section 2, premise P3)	**Wootters (1981)** introduces the statistical distance between states. **Caticha (2005, 2016, 2019)** proposes that the metric of space is the information metric of a point with a blurred position. This fixes the shape of space (conformal geometry) but leaves the local scale ψ free (2016). Caticha uses the blur as the unit and the geometric conditions	The starting statistical object is not the blurred position of a point. It is a **register of counts**, with its Dirichlet distribution. The **three dimensions** of space come from the fact that the categories are **K = 4**.

	of Hojman, Kuchař and Teitelboim. In this way he obtains Einstein's vacuum equations (2019).	
Premise P1: proper time advances with the size of a fluctuation	**Entropic time** of Caticha (2011). **Local entropic time** of Ipek, Abedi and Caticha (2018) and of Ipek and Caticha (2020). At each point the variance of the fluctuations grows by equal amounts in equal intervals of proper time: $\delta\xi\perp = N\,dt$.	In their works the rate N is already given by the geometry. It is N that sets how large the fluctuations are. **Here the arrow is reversed.** The step for each reading depends on the density of the counts: $\sqrt{(K-1)}/\tau$. And it is the ratio between the steps of two places that gives the rate N.
$N = \tau_0/\tau_x$, the rate as a ratio between far and near	**Tolman and Ehrenfest (1930):** in a fluid in thermal equilibrium in gravity $T_{loc}\,N = T_\infty$ holds, that is, $N = T_\infty / T_{loc}$.	In place of the temperature there is another quantity: the **density of readings τ** of the register.
ψ, the conformal factor of the metric of space	**Caticha (2016)** notes that ψ is the local scale left free by information geometry. In general relativity it is then fixed by the Lichnerowicz–York equation. In time-symmetric vacuum this reduces to $\nabla^2\psi = 0$.	Here **ψ is read directly as a share** of a joint count (3.4-3.5). With Postulate T it becomes $\psi = 1 + \Sigma_i\, Gm_i/(2r_ic^2)$. It is the Brill–Lindquist solution of that equation.
Postulate T: the quantity w (equal to $-\Phi/2c^2$ at first order) as a share between 0 and 1	**Verlinde** reads $-\Phi/2c^2$ as the degree of loss of detail (*coarse graining*) per bit. It is between 0 and 1 and equals 1 only on the horizon (preprint of 2010). In the published version of 2011 he writes $-2\Phi/c^2$. This equals 1 on the horizon in areal coordinates.	Here the **completeness of the two shares** is added, $\psi + N\psi = 2$. It ties the rate N rigidly to the spatial factor ψ. And it produces the exact isotropic Schwarzschild metric, at all orders.
Counting degrees of freedom and gravity as thermodynamics	**Padmanabhan (2010)** links energy to the degrees of freedom on the horizon: $E = \tfrac{1}{2}\, k_B \int dn\, T_{loc}$, with $dn = \sqrt{\sigma}\, d^2x/L_P^2$ and $T_{loc} = N\, a\cdot n/2\pi$. **Jacobson (1995, 2016)** obtains Einstein's equation as a thermodynamic equation of state. He also obtains it from the equilibrium of quantum entanglement.	In those works there is no direct reading of the numbers N and ψ as ratios and shares of elementary counts.

In the works listed above we did not find four elements. They are the specific proposal of this deposit, always within the limits declared in sections 3.9, 5 and 8:

1. the rate of time N read as a ratio of densities of readings, τ_0/τ_x, starting from the Fisher step $\sqrt{(K-1)}/\tau$;
2. the spatial depth ψ read as the directional overlap of the projected images, $2\tau_x/(\tau_x + \tau_0)$, with the rule $\psi + N\psi = 2$ understood as the completeness of two shares that sum to one;
3. the fragmentation of a 4-category register, which behaves as a Brownian motion and deposits the 1/r law;
4. the phenomenological closure of the equations inside matter (section 4).

This check covered the works listed in the bibliography. It did not cover the whole world literature on gravity.

10. A clock in motion (outlook; added in version 2.36.1)

Now let us leave gravity. Let us see how the counting of copies gives a reason for the slowing down of fast clocks, in flat space.

Section 3 talked about clocks at rest near a mass. Let us take instead a clock that moves at constant speed. It moves in a flat spacetime, without gravity. Here the reading based on counts produces the **time dilation** of Einstein's special relativity.

Imagine the start event and the arrival event of the clock. Between these two points, the light rays draw a four-sided figure. It is called a **light diamond**, or causal diamond. The two sides of the diamond along the directions of light are **$a = c\Delta t - \Delta x$** and **$b = c\Delta t + \Delta x$**. Here Δt is the time passed in the laboratory. Δx is the distance travelled by the clock.

Here two more premises of the MM framework come into play.

- **P6 (second link).** In an elementary step of the motion, the two counts along the light sides are **coprime**. Two integers are coprime if they have no common divisors other than 1, like 3 and 5. The copies are born at their common multiples. Therefore a diamond with sides a and b contains a number of cells equal to the product **$ab = \gcd(a, b)\cdot\mathrm{lcm}(a, b)$**. Here gcd is the greatest common divisor and lcm the least common multiple.
- **P7 (no place assigned in advance).** Places in space do not exist before the copies arrive. A grid of places fixed in advance would define a privileged frame of reference: a motionless *ether*. In the computer simulations the copies are therefore distributed independently and at random in the volume of spacetime. It is a practical assumption. Premise P7 motivates it, but does not impose it in a mathematical sense.

[S] Declared choice. The moving clock reads the **number** of copies inside the diamond. It combines the two opposite directions of light by multiplying them together: this is the product rule. The time marked by the moving clock is thus the square root of the product: **$\sqrt{(ab)}$**. The time of the clocks at rest in the laboratory is instead the average of the two sides: **$(a + b)/2$**.

[C] Test results. We tried several arrangements of the copies. And two ways of reading the clock. The first counts the *longest chain* of successive steps, from the beginning to the end. The second uses the product rule directly. Here are the outcomes:

Arrangement of the copies in spacetime	What the clock reads	Outcome of the calculation
Random, uniform, in 1+1 dimensions (1 of space + 1 of time), 200,000 copies	Longest chain	It gives Einstein's factor **$\sqrt{(1 - v^2/c^2)}$**. The deviation is between **+0.2 % and −0.9 %** for speeds from v = 0.3 to v = 0.9 (in units of c). It is within **0.3 %** after correcting for the effect of the finite number of copies.
Random, uniform, in 3+1 dimensions (3 of space + 1 of time), 6000 copies	Longest chain	A moving clock and one at rest with the same proper time agree within **1.3 σ** (standard deviations). But at v = 0.8 **the fixed tolerance threshold of 6 %, registered before the test, is not met**. There only about 780 copies remain in the diamond.
Synchronous grid (points already aligned in space and time)	Longest chain	**The reading falls as a reading of relativity.** It gives **$(a + b)/2$** at every speed v. It is **Galileo's absolute time**, with no dilation at all.
Fixed places in space, random times	Longest chain	**The reading falls.** It gives a slowing down **from 7 % to 29 % stronger** than the true Lorentz one. The fixed places create a privileged frame of reference.
Synchronous grid	Number of copies with the product rule	It gives **$\sqrt{(1 - v^2/c^2)}$** within **0.43 %** (for T = 1000).
Lattice along the light cones	Direct comparison of $\sqrt{(ab)}$ with $(a + b)/2$	It reproduces **$\sqrt{(1 - v^2/c^2)}$** with the extreme precision of the machine, down to **3×10^{-16}**.

The mathematical reason for all this fits in one line. The area of the diamond is proportional to the product of the sides **ab**. So is the number of copies that fall inside it. Therefore a clock that counts the copies and takes the square root, as the product rule requires, measures the **geometric mean** of the two sides: **$\sqrt{(ab)}$**. The same holds for a clock that measures the longest chain, because the chain grows as the square root of the count. A laboratory clock adds the two sides. It measures instead the **arithmetic mean**: **$(a + b)/2$**.

In mathematics there is a very simple rule. **The arithmetic mean is always larger than the geometric mean.** The only exception is the case of two equal sides (a = b), that is, the clock at rest. **And the ratio between the two means is exactly Einstein's time-dilation factor.** An example with numbers chosen by us: a = 2 and b = 8. The arithmetic mean is $(2 + 8)/2 = 5$. The geometric mean is $\sqrt{(2 \times 8)} = \sqrt{16} = 4$. The moving clock marks 4, while 5 pass in the laboratory. With formulas:

$\sqrt{(ab)} / ((a + b)/2) = \sqrt{(1 - v^2/c^2)}$

Let us see how the clock advances, step by step, on the light lattice. A clock at rest makes symmetric steps of **(1, 1)**. A step with coprime sides a and b is **indivisible**. The straight segment from (0, 0) to (a, b) contains inside it exactly **gcd(a, b) − 1** points of the grid. When the greatest common divisor is 1, the interior points are zero. This was checked without any exception on **3600** pairs of numbers. A moving clock advances in space by **$(b - a)/2$** at each elementary step.

Note on the registered outcome. Before running tests M1 to M5 a strict overall criterion had been fixed. If even just one of M1 and M5 had missed its threshold, the dilation would not have been considered verified by the count of the chain. This overall criterion **was not met**. Test M5 in 3+1 dimensions went beyond the fixed tolerance of 6 % at the speed v = 0.8. The reason is the small number of copies left in the narrow diamond: about 780. The comparison at equal proper time, that is, with the same number of copies, passed instead. The analysis of the finite-size effect was made after seeing the results.

Earlier work on the moving clock

From the mathematical point of view, none of these geometric results is new.

- Computing proper time as the longest chain of randomly scattered points (Poisson distribution, or *Poisson*

sprinkling) is a classical result. So is proving that it respects Lorentz relativity. It comes from causal-set theory: **Brightwell and Gregory (1991)**; **Bombelli, Henson and Sorkin (2009)**.
- In 1+1 dimensions, finding the length of the longest chain is a famous problem: **Ulam's problem**. It was solved by **Hammersley 1972**, **Vershik and Kerov 1977**, **Logan and Shepp 1977**.
- Writing proper time as the geometric mean √(ab) of the two light times is the heart of **Bondi's k-calculus (1964)**.

What, then, does the MM framework add? It gives a **physical reason** why the copies must be arranged in that way. Premise P7 forbids a fixed grid of places, which would give Galileo's time. Premise P6 requires that the count combines the two sides with a product.

What is not done in this section

- **Joining the two clocks.** This count of the moving clock is not yet joined in a single chain with the gravitational rate N = τ_0/τ_x of section 3. In this deposit the slowing down due to gravity and the one due to speed are computed with two separate counts.
- **Curved spacetime** for the moving clock.
- **Positions of the copies linked to each other**, that is, not independent.

The continuation of this research on the clock will be a separate deposit, in preparation. Weights are put on the changes of direction. In this way one gets Feynman's checkerboard and the Dirac equation for the electron. There mass appears as the turning angle. On the same lattice one also finds again the field studied in MM 2.27.

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11. The relation as an exchange and the deposit as a probability of arrival (added in version 2.36.2)

In this section we take two steps. First: we read the rule ψ + Nψ = 2 as a continuous exchange. The exchange protects the unit of measurement of the distant observer. Second: we read the transferred share w as a probability of arrival.

11.1 What the relation fixes at second order [C]

Take ψ = 1 + w and w = GM/(2rc²) from Postulate T (section 3.7). Let us look at the quantity **Nψ²**. We write it as a function of w: **Nψ² = F(w)**. What is Nψ² in physics? N is the local unit of time. ψ² is the local unit of length. Both are compared with those of the distant observer. So **Nψ² is the product of the local unit of time and the local unit of length**.

Let us try three different forms for the function F(w). For each one we look at what happens to the orbits of the planets and to light. We always read the mass from the behaviour of g_tt at large distance, that is, at first order. And we use the standard variable U = GM/(rc²).

Choice for F(w) = Nψ²	What −g_tt = N² becomes	Parameter γ	Parameter β	Outcome of the comparison with observations
F(w) = 1	1 − 2U + 5U²/2	1	5/4	**The reading falls (excluded).** It gives β = 5/4, that is, a quarter more than 1. The motion of the perihelion of the planets contradicts it.
F(w) = 1 − w	1 − 3U + 19U²/4 (after rescaling the mass)	2/3	19/18	**The reading falls (excluded).** It gives γ = 2/3. The deflection of light near the Sun and the Shapiro radar delay contradict it.
F(w) = 1 − w²	1 − 2U + 2U²	1	1	**Exact isotropic Schwarzschild**, with N = (1 − w)/(1 + w).

In general we expand F as a power series: F = 1 + f_1w + f_2w² + … The calculations give γ = 1/(1 − f_1/2). And, when f_1 = 0, they give β = (5 + f_2)/4. The Schwarzschild values are **γ = 1 and β = 1**. Experiments confirm them with great precision (Will 2014; Bertotti, Iess and Tortora 2003 for γ). To have this agreement two things are needed. **f_1 = 0** is needed: no term linear in w. And **f_2 = −1** is needed: the squared term must be exactly **−w²**.

Now let us look at our relation **ψ + Nψ = 2**. Since ψ = 1 + w, the relation says that Nψ = 2 − ψ = 1 − w. Let us multiply ψ by Nψ. We get: Nψ² = ψ · (Nψ) = (1 + w)(1 − w) = **1 − w²**. The relation ψ + Nψ = 2 is therefore exactly equivalent to **F(w) = 1 − w²**. And after −w² there are no other terms, at any order. This is the content of choice S4 of section 3.7.

The product $N\psi^2$ also has a direct geometric meaning. We call $R = r\psi^2$ the areal radius. It is the radius for which the surface of the sphere is $4\pi R^2$. Its derivative with respect to the isotropic radius r is exactly $\mathbf{dR/dr = 1 - w^2 = N\psi^2}$.

11.2 The reading as an exchange [P, S, C]

Why in physics should exactly the product $(1 + w)(1 - w) = 1 - w^2$ appear? Here is the author's premise that gives a reason for it.

[P] Premise P8 (exchange). Light is a change in the configuration of the description. The unit of reading of the reference observer, the distant one, is stable. To keep it stable, the fractional parts of the description continuously exchange quanta. Quanta are elementary packets: fractions of that unit.

[S] Declared formalisation. We translate premise P8 into formulas with three identifications.

1. The two parts that exchange fractions are the two shares of section 3.5: $\mathbf{\psi/2 = (1 + w)/2}$ and $\mathbf{N\psi/2 = (1 - w)/2}$. Each one moves by w/2, in opposite directions. One grows by w/2, the other drops by w/2.
2. The unit of the distant observer is stable: this means that the sum of the two shares stays constant. $\mathbf{\psi + N\psi = 2.}$
3. Light is a change of configuration. It reads the **product** of the local units of time and length: $\mathbf{N\psi^2 = \psi \cdot N\psi}$.

[C] Consequence. From these three lines $\mathbf{N\psi^2 = (1 + w)(1 - w) = 1 - w^2}$ follows at once. The term $-w^2$, the one required by the observations in section 11.1, arises like this: from a fixed sum read through a product. In particular **it is not assumed that the two shares are statistically independent**. In probability two events multiply when they are independent. Here instead the two shares are dependent to the highest degree: what one gives is exactly what the other receives.

There is more. The ratio between the two shares, $\mathbf{\psi / (N\psi) = 1 / N}$, coincides exactly with $\mathbf{1 + z}$. It is the gravitational shift towards the red (*redshift*) of light emitted at the place and received far away. The number **w** measures the imbalance between the two shares. It is twice the transferred fraction w/2. We see it in the next paragraph: in the ideal Brownian model w is also a probability of arrival.

Logical status. P8 is a premise. The three identifications are a declared translation of it. As in 3.5, the two shares sum to one by construction. P8 reads this fact as an exchange. But here it does not give an equation of motion, that is, a dynamics of how the exchange happens in time. If we keep this reading point by point inside a star, we find again the closure of section 4.

11.3 The deposit read as a probability of arrival [C]

Now let us see a link between gravity and the probability that a random walk hits a target.

Take a Brownian motion in three-dimensional space. It walks without ever stopping. It starts at distance r from a small target sphere of radius ε (with $r \geq \varepsilon$). What is the probability that sooner or later it touches that small sphere? A classical theorem answers: Kakutani 1944, valid for regular targets in $\mathbb{R}^3$. This **probability of arrival** is exactly $\mathbf{\varepsilon / r}$.

Let us compare this formula with our deposit of section 3.6, which was $1/(2\pi r)$. The two numbers differ only by a fixed constant: $2\pi\varepsilon$. Now let us look at Postulate T (section 3.7). It says that the share moved is $w = GM/(2rc^2)$. When does the share w coincide exactly with the probability of arrival ε/r, for every distance r? When the radius of the target is:

$\varepsilon = GM / (2c^2)$. It is exactly the radius of the black-hole horizon in isotropic coordinates.

The coefficient of Postulate T is therefore exactly the radius of the isotropic horizon. For the Sun it is about **738 m**. For the Earth it is **2.2 mm**. They are virtual targets at the centre of the body. Let us take flat geometry as a reference (Euclidean metric δ). With respect to it the function $w = \varepsilon/r$ is harmonic. It is 1 on the sphere of the horizon $r = GM/(2c^2)$. It falls to 0 at infinity. So w is the exact probability that a Brownian path in flat space reaches the sphere of the horizon.

And what if the Brownian motion walks in the **curved geometry** of the space of the star or of the black hole? This geometry is described by the metric $\mathbf{\psi^4\delta}$. In that curved space the operator of Brownian motion is $\frac{1}{2}\Delta_g$. There the function w is no longer harmonic: in fact $\Delta_g w = 2\psi^{-5}|\nabla w|^2$, which is not zero. What, then, is the harmonic function that is 1 on the horizon and 0 at infinity, in the curved geometry? The calculation gives $\mathbf{1 - N = 2w/(1 + w)}$.

$1 - N$ is the probability of falling into the horizon. Its complement to one, that is, **N**, then has a clear meaning: **the rate of clocks N (the lapse) is the probability that a Brownian path, in the curved geometry of space, escapes to infinity without ever touching the horizon.**

In general relativity it is known that the lapse N of the static vacuum is a harmonic function in the metric of space. It is a deep mathematical property. It is used for example to prove the uniqueness theorem for black holes (Bunting and Masood-ul-Alam 1987). And it defines the *capacity* of the horizon in Bray's famous mass–capacity inequality (2001). In short:

- in the **Euclidean reading**, that is, the flat one, the imbalance **w** is the probability of arriving at the horizon;
- in the **curved reading of space**, **N** is the probability of escaping to infinity and **1 − N** is the probability of arriving at the horizon.

These statements hold in the empty exterior space ($r \geq GM/(2c^2)$). The horizon is treated as an absorbing boundary. And the path has an infinite time available. For normal bodies, like the Sun or the Earth, the horizon would be buried inside the matter. There the target is only virtual. The reading describes the idealised exterior field.

11.4 The test on the fragmentation test bench [C]

This link between deposit and probability of arrival holds for a continuous and ideal Brownian motion. Does it also hold for the discrete steps of our **fragmentation of registers** (section 3.6)?

The author checked this with a series of computer tests. The predictions were registered before the calculations. At the original scale of the test bench of section 3.6 a geometric problem emerged at once. At distance r from the source a chain has about $1/(r^2 + 1/T)$ readings. So it makes steps of a length similar to $\mathbf{r^2}$. These steps are not small compared with the target cell used in 3.6. The chain risks *jumping over* the cell without touching it. It is like a giant who steps over a pebble without treading on it.

To reduce this error due to the discrete steps, the whole geometry of the test was **scaled down 4 times** (scale 1/4). And every single reading was followed, without skipping any. There are **4,000,000 readings** for each chain, on **3000 chains** simulated. The table gives the ratio between the measured probability of arrival and the theoretical Brownian law, at six distances r. A ratio equal to 1 would mean perfect agreement.

Radius r (in the bench scaled down by 1/4)	**0.0015**	**0.002**	**0.00275**	**0.00375**	**0.005**	**0.00675**
Probability of arrival / Brownian law (original scale)	0.925	0.900	0.857	0.809	0.732	0.658
The same, scaled down by 1/4 following the exact path	**0.998**	**0.993**	**0.976**	**0.952**	**0.915**	**0.904**
Statistical error of the measurement	0.011	0.012	0.013	0.014	0.014	0.014

Here are all the outcomes: the positive ones and the registered thresholds that were not met.

- **Proportionality to the deposit.** On the bench scaled down by 1/4 the probability of arrival is proportional to the deposit within **4.4 %**. It is the largest difference between the central values, with the registered estimator. The proportionality constant is $0.952 \times 2\pi\varepsilon$. The direct comparison on the same shell, made after the calculation, gives **4.5 %** and constant **0.948**.
- **Reduction of the deficit.** At the two smallest radii (0.0015 and 0.002) the difference from 1 is almost zero: 0.998 and 0.993. It is within the statistical error of 0.011 and 0.012. At the other radii the deficit with respect to 1 has shrunk **from 3 to 6 times** compared with the original scale. This confirms one thing: a large part of the arrivals missed at the original scale depended on the size of the discrete steps.
- **Arrivals lost by looking in blocks.** Instead of checking every reading, we can check the position only in blocks of readings. How many arrivals do we lose this way? The chain can enter and leave the cell between one block and the next. Measured on the same chains, **4.7 %** of the arrivals are lost at the smallest radius. At the largest radius **1.3 %** is lost.
- **Comparison with the predictions registered before the test.** A registered prediction said by how much the deviation would shrink when moving to scale 1/4. This prediction **held**. Instead the strictest threshold, E3, **was not met**. It asked to stay above **0.95** at all radii. At the two largest radii the value stopped at **0.915 and 0.904**. Then there is check E1, on the fast block generator compared with the exact path. It narrowly missed its tolerance. And in the first tests, at the original scale, the registered criteria had not been met.

In conclusion, the numerical results **support the hypothesis** that in the continuous limit the first arrival of fragmentation follows the law of Brownian motion. This is what the mathematical invariance principle leads one to expect. For a rigorous proof, however, a convergence theorem made specifically for the paths of fragmentation would be needed. Let us remember that this whole bench lives in the Fisher space of register states (premise P3). And it stops at a finite

maximum time, s_max. This is where the erfc correction factors come from.

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12. Answers to the most frequent questions (anticipated questions)

Here are seven critical questions that a careful reader might ask. Each one has its direct answer.

1. **Between which distributions is the overlap computed? And why don't you use the relative entropy?**

It is computed between the Dirichlet distributions of the place and of the distant observer. They are projected on the plane perpendicular to the line of sight that joins them (section 3.4). With many readings these spots are round (isotropic) and have width $1/\tau$. We use the normalised overlap because its two opposite directions sum to a fixed constant: 2 in the plane. With the relative entropy, instead, the sum changes from point to point (section 3.1). Taking the direction seen from the place is a declared choice.

1. **Outside matter, isn't the rule $\psi + N\psi = 2$ just a trivial consequence? Laplace's equation in vacuum is linear.**

Yes. Outside matter the relation is a simple property of general relativity in vacuum. The new physical content of the model is **inside matter** (section 4).

1. **What is the general covariant form, that is, valid in any system of coordinates? From which quantities are ψ and N read in general?**

For now **there is no general covariant form**. However, in our restricted domain, static and spherically symmetric, ψ and N are defined without ambiguity. N is the length (norm) of the static Killing vector, which indicates the flow of static time. ψ^4 is the scale factor of the metric of space. In fact every three-dimensional space with spherical symmetry can be written as a flat grid multiplied by a factor ψ^4.

1. **Why do you use D = 2 dimensions in the overlap formula, if physical space has three?**

Because the comparison is made on the **image** projected on a screen perpendicular to the line of sight (premise P2). In the whole three-dimensional state space the exponent 3/2 appears. Then the sum is no longer 2 (section 3.4). The work does not identify a laboratory instrument that measures this overlap directly.

1. **Which Einstein equation is dropped inside a star? And what happens to the energy conditions?**

The **lapse equation** of general relativity is dropped. The Hamiltonian constraint is kept. The pressure is obtained from hydrostatic equilibrium, that is, from conservation. For this reason the pressure of the fluid no longer generates a gravitational field. Seen through the glasses of general relativity, the star has an extra anisotropic stress. If the compactness exceeds 4/9, this stress must violate one of the conditions of Andréasson (2008): $p_r \geq 0$ or $p_r + 2p_t \leq \rho$. In the uniform-density star the pressure of the fluid stays positive anyway. And it falls smoothly from the centre to the surface. The energy conditions are studied in detail in the deposit MM 2.34.

1. **Why, in Postulate T, do you choose normalisation S4 and not S3?**

Both pass today's tests in the Solar System, because they give $\gamma = \beta = 1$. But S4 reproduces exactly the Schwarzschild metric at all orders. S3 instead deviates from it at third order. The choice of S4 uses general relativity as an external guiding criterion. It is not imposed by the premises of MM.

1. **Isn't the reasoning circular? That is: do you put Schwarzschild in at the beginning to find it again at the end?**

The point where Schwarzschild enters is openly declared. It enters with Postulate T and choice S4 (section 3.7). All the other steps of the chain compute mathematical forms: a ratio of densities, a share between two points of view, a 1/r deposit from a Brownian motion. These forms do not contain the metric of gravity.

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13. How the work was done

This section collects the information on the working method and on the origin of the calculations. It also says how anyone can check and redo every calculation on a computer. The archived text ends with a *Sintesi in italiano* (summary in Italian), in 11 points: its contents are all already in sections 1-12 of this version.

Provenance and method of checking

The author is not a professional physicist. The MM theoretical framework and its premises, including premise P8, are his. The calculations and the programs were developed with the help of artificial intelligence assistants (Claude and Codex).

The numbered tests had to be honest: no changing the rules after seeing the results. For this reason their predictions were registered before running the programs. For each prediction a cryptographic fingerprint (**hash**) was computed. The fingerprints are in the author's private working files. They do not have a public time stamp. Every check or deeper study made *after* seeing the results is declared as such in the text.

Some results were redone independently by a second AI assistant, not by a human referee. In particular:

- The chain of section 3 came from a numbered series of registered tests (described in the file `documenti/TUNNEL_TESTS.md`). The second AI assistant checked it in five rounds. All its corrections were applied.
- Version 2.36.1 (October 2026) added section 9, after a literature check made with the same AI assistant. It also added section 10 on the moving clock. This came from a separate series of registered tests (M1-M7, described in `documenti/MOVING_CLOCK_TESTS.md`), with three dedicated programs. Every other program, result and log of version 2.36 stayed the same.
- Version 2.36.2 (October 2026) added section 11. It comes from another series of registered tests on the fragmentation bench (`documenti/ARRIVAL_TESTS.md`). The registrations and the original notes, in Italian, are in the folder `documenti/registrations_arrival/`. The nine programs were checked by the second assistant before the deposit.
- Before publishing the deposit on Zenodo, the whole package of programs was run again from scratch. Every result was reproduced in full.

How to redo all the calculations (Python 3 with the libraries numpy, scipy, sympy)

All the programs are in the folder `codice/`. The official reference results are in the folder `risultati/`. When you run the programs on your computer, the new files go into the folder `uscite/`. The SHA-256 fingerprints of all the files are in `IMPRONTE.sha256`: they are used to check that nothing has been altered. The comments in the programs and the messages on screen are in Italian, or partly in Italian.

Here are the main commands to type in the terminal.

- `python codice/run_all.py` runs all the programs listed below, including the three of section 10 and the nine of section 11. It saves the logs and the JSON files in `uscite/` and compares them with `risultati/`. It does not look at differences in line endings. It hides the seconds taken by the programs `grafo_uno_su_r.py` and `angolo_retto.py`. If even a single number differs or is missing, it reports an error. It takes about **90 minutes**.
- `python codice/run_all.py --short` does everything except the three longest calculations of section 11. It takes about **25 minutes**.
- `python codice/ppn_quadrato.py`, for section 11.1: computes γ and β for F = 1, 1 − w, 1 − w².
- `python codice/einstein_vuoto_esatto.py`, for sections 1 and 3.8: checks that with a single centre one gets isotropic Schwarzschild, with the Einstein tensor equal to zero.
- `python codice/domini_e_masse.py`, for sections 1, 3.8 and 4: checks that two centres are not a static solution in vacuum. It computes the flux 4π(M_Komar − M_ADM).
- `python codice/specchio.py` checks ψ + Nψ = 2 in vacuum. And it checks that in general relativity, inside matter, the relation does not hold point by point.
- `python codice/interno_uniforme.py`, for sections 4 and 12: computes the uniform-density star, C_max in closed form, the central pressure and the extra anisotropic stress.
- `python codice/reciprocita_nello_specchio.py`, for section 3.1: checks the identity of the Gaussian overlaps and the comparison with the relative entropy KL.
- `python codice/perielio.py`, for section 3.8: checks β = γ = 1 and computes the integrated exterior orbits.
- `python codice/gamma_anello.py`, for section 7: computes γ on a ring with finite counts (notes in `documenti/NOTE_GAMMA.md`).
- `python codice/ritmo_per_lettura.py`, for section 3.3: computes the expected Fisher step for each new count

(notes in `documenti/NOTE_FISHER_STEP.md`).

- `python codice/normalizzazioni.py`, for section 3.7: compares the four normalisations S1-S4.
- `python codice/grafo_uno_su_r.py [n N seed]`, for sections 2 and 6: computes the free exponent of the response on the graph. By default it uses n = 512, which runs on a laptop in about a minute. The cases n = 1024 and n = 2048 need more than **8 GB** of RAM memory. The case n = 2048 took **135 s** on a machine with **128 GB** of RAM.

The programs of the registered tests write their JSON files in the folder you are in. So they must be launched from inside `uscite/`, as `run_all.py` does by itself.

- **For section 3:** `t1_densita.py` and `t5_coerenza.py` (3.3); `t6_specchio.py`, `t9_immagine_esatta.py` and `t9c_controllo.py` (3.4, including the check of the estimator on discs and Gaussians); `t8b_frazionamento.py` (3.6, the longest calculation of section 3); `t11_due_masse.py` and `t11b_tensore.py` (3.8).
- **For section 10 (moving clock):** `angolo_retto.py` runs the tests M1-M5, with copies at random, on a grid and at fixed places, in 1+1 and 3+1 dimensions, in about **45 s**. Then `conteggio_p6.py` (test M6: count with the product rule) and `orologio_fermo_p7.py` (test M7: indivisible steps and clock at rest).
- **For section 11 (arrival and exchange):** `cella.py` (first test at the original scale, on **6000** chains); `cella_diagnostica_eps.py 0.004|0.008` (diagnostic test with larger cells, made afterwards); `scala.py A|B 3` (block generator); `valuta_scala.py`; `diag_griglia_A.py 1000` (measurement of the arrivals lost by looking only at the block points); `scala_esatta.py 3000 3` (follows the exact path reading by reading on 3000 chains, in about **45 minutes** with three processor cores); `valuta_scala_esatta.py`; finally `quota_e_arrivo.py` (symbolic calculation for 11.2-11.3, which also reads the data of `scala_esatta.json`).

This English text is a translation of the Italian student version, which the author read and approved.

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14. Glossary of key words

Term	What it means in this work
Bayesian overlap	The number that measures how much two registers, or two distributions, resemble each other and overlap: w_ij in section 2, $\ell(a \to b)$ in section 3 for the Gaussian curves.
Copy	In section 10 on the moving clock, a single elementary event of the count. By premise P7 the copies have no places fixed in advance.
Deposit	The trace left at distance r by the fragmentation, when each step is weighted with the precision lost Δs. It follows the law $1/(2\pi r)$.
Depth ψ (conformal factor)	The number that says how much the scale of lengths changes at a place, compared with the distant observer. In MM it is read as the directional overlap $2\tau_x/(\tau_x + \tau_0)$.
Distant observer	The reference point of view, very far from the mass, where there is no gravitational field. It has density of readings τ_0, depth $\psi = 1$ and rate $N = 1$.
Effective density τ	The total number of readings of a register, plus one: $\tau = T + 1$.
Exchange	In section 11, the continuous passage of fractions of the distant observer's unit of measurement between the two shares (premise P8). It keeps their sum fixed: $\psi + N\psi = 2$.
Fragmentation	The process by which the copies of a register lose precision, reading by reading. In the Fisher space with 3 dimensions it behaves as a Brownian motion.
Light sides a, b	The two sides along the light rays that close the diamond between the start and the arrival of a clock. They are $c\Delta t - \Delta x$ and $c\Delta t + \Delta x$.
MM (Mathematical Materiality)	The author's theoretical framework. Here physical quantities arise from the statistical comparison between lists of counts.
Postulate T	The transfer law of section 3.7. It links the geometric deposit to physical gravity, moving a share $w = GM/(2rc^2)$.

Premise	A basic principle (axiom, from P1 to P8). It is the starting point of the reasoning and it is not proved.
Probability of arrival	The probability that a path of the fragmentation, or a Brownian motion, reaches a given target cell or the black-hole horizon.
Rate N (lapse)	The number that says how slow a clock runs at a place, compared with the distant observer. In MM it is the ratio of the densities of readings τ_0/τ_x.
Reading	A single count that is added to a register.
Register	A list (vector) of counts divided into K = 4 categories, with its Dirichlet probability distribution.
Share	The fraction ψ/2 or Nψ/2. It says which part of the joint total count belongs to the observed place and which to the distant observer.

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